LINFENG WANG, PhD

wanglinfeng1115@gmail.com ⋄ +44 7599610897 ⋄ linkedin.com/in/w15 ⋄ linfeng-wang.github.io/

Skills

Language & Frameworks:	**Python, PyTorch**, PyG, R, Bash, MySQL, HTML/CSS, C++
Machine learning:	Scikit-learn, Transformers, LLMs, CNNs, RNNs, GNNs, VAEs, XGBoost
Development Tool:	AWS, GCP, Docker, Git, Nextflow, Pandas, NumPy, Jupyter, Huggingface, LangChain, Numba
Visualization:	Matplotlib, Plotly, ggplot2, streamlit, Flask

Experiences

Machiner learning engineer - Biographica, London UK **Apr 2026 –**

- Develop, tested and benchmark **deep learning models** predicting gene expression from **DNA sequence and biological context**.
- Developed scalable **PyTorch training and fine-tuning pipelines** on **Azure and AWS infrastructure**, including dataset construction, model evaluation, hyperparameter optimisation and reproducible experimentation.
- Applied ML to **gene-editing and crop-trait prediction**, combining multi-omics data

PhD researcher - LSHTM (Clark Campino Phelan lab), London UK **Oct 2021 – Mar 2026**

- Developed **dockerised** reproducible deep learning models (**CNN, RNN, GNN, Transformer**) with **PyTorch**
- Built **statistical pipelines** on **HPC** in Python for GWAS, PCA, logistic regression, and odds ratio analysis on **high dimensional data**
- Developed a **web application** and **backend software (PyPi package)** in **Python** for automating the design of amplicon primers.
- Developed an **RNN-based generative NLP model** for **novel antimicrobial peptide sequence design & classification with 91% accuracy**
- Developed an **GNN-based classification model** for natural selection prediction based on phylogenetic trees achieving **88% accuracy**
- Built and deployed a **RAG LLM** chatbot integrating **ChromaDB**, **LangChain**, and OpenAI API for course content and feedback collection

Machine learning consultant - Deep Science Venture, London UK **Mar 2025 – Oct 2025**

Project: Sequence-Based Drug Discovery using Deep Generative Models

- Led **CNN, RNN, and VAE** models developments on biological sequences with data augmentation and hyperparameter optimization
- Producing a model of **86% accuracy** in predicting gene expression
- Applied **SHAP, LIME, and DeepLIFT** to extract interpretable biological insights from deep models
- Directed strategic model design for **generative sequence discovery** tasks

Data study group hackathon participant – The Alan Turing Institute, London UK. **Jan 2024**

Project: Shallow Gas Hazard Detection in Offshore Seismic Data – led to Turing publication see publication section

- Led deep learning model development for detection and segmentation on geophysical imagery using **CNNs with contrastive learning**
- Delivered **93% classification accuracy** on legacy seismic datasets under 2-weeks
- Collaborated across domain boundaries to translate geoscience challenges **into ML solutions**

Machine learning Intern - Linkgevity, London UK **Aug 2024 – Oct 2024**

Project: Graph-Based Drug-Drug Interaction Prediction

- Recreated **GNN** models with with integration of **BERT** to predict drug compound interactions
- Trained **drug-drug interaction model** for **drug design**, **improved predictive performance by 15%**
- Scaled ML training workflows using **Google Cloud Platform** and **AWS** for high-throughput experimentation

Data Science intern - ByteDance, London UK **Aug 2022 – Feb 2023**

Project: Neoantigen Insight Platform and Market Landscape Scouting

- Built **SQL-integrated chemical database** for drug candidates
- Used **autoencoders** to uncover hidden patterns in immune-related molecules across biological datasets
- Coordinated **interdisciplinary market research** on biologics and small molecules
- Authored bi-weekly **research briefings** and developed end-to-end ML pipelines for **drug discovery.**

SELECTED PUBLICATIONS (13 in total, 7 First authored expected)

- Antinucci, **Wang et al.** Scaling a **context-aware genomic model** for condition-specific edit design in plants. preprint: https://doi.org/10.64898/2026.10.23.761389
- **Wang et al.** TOAST: A novel tool for designing targeted gene amplicons. *BMC genomics*. **doi:** https://doi.org/10.1186/s12864-025-12247-9 - **Website:** https://genomics.lshtm.ac.uk/webtoast/#/ - **Pypi software package**: https://pypi.org/project/toast-amplicon/
- **Wang et al.** LSTM-Based Models for Antimicrobial Peptide Classification and Generation. **doi:** https://doi.org/10.1093/bioadv/vbaf274
- **Wang et al.** Decoding Positive Selection with Phylogeny-Guided **Graph Attention Models**. **doi**: https://link.springer.com/article/10.1186/s12859-026-06583-0
- **Wang et al.** ML for stepwise prediction of TB treatment outcomes. preprint: https://doi.org/10.21203/rs.3.rs-7558046/v1
- DSG Report: British Geological Survey - Detecting Shallow Gas from Marine Seismic Images" The Alan Turing Institute (2025) **link:** https://www.turing.ac.uk/news/publications/data-study-group-final-report-british-geological-survey-0

EDUCATION

UCL, London School of Hygiene and Tropical Medicine , DPhil Computational Genomics **Oct 2021 – Oct 2025**

Dissertation: Integrative Genomic Sequencing and Machine Learning for TB Drug resistance, Diagnostic Tool Development
Award: UKRI – BBSRC – LiDo PhD scholarship

Imperial College London, MRes Bioengineering (Hons), Merit **Oct 2019 – Oct 2020**
Relevant courses: *Data Analysis for Research*

King's College London, BSc Biochemistry (Hons), 1st **Sep 2016 – Jul 2019**
Relevant courses: Bioinformatics, Protein design and functions, Molecular dynamic modelling
Awards: Wellcome Trust Biomedical Vacational Studentship (2018)